

HDOC Day-36 - C Programming

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Problem Statement

Write a menu-driven C program to perform the following operations using separate functions:

1. Count characters in a string without using built-in length functions.
2. Print each character of a string on a new line.
3. Count vowels and consonants in a string.
4. Exit.

Algorithm

1. Start the program and display the menu.
2. For option 1, read a string and increment a counter until the null character is reached.
3. For option 2, read a string and print each character on a separate line.
4. For option 3, read a string, examine each alphabetic character, and count vowels and consonants separately.
5. Repeat the menu until option 4 is selected.
6. Stop.

Test Case Table

Choice	Input	Expected Result
1	Dhruv	Number of characters = 5
2	Dhruv	D, h, r, u, v printed on separate lines
3	Dhruv	Vowels = 1, Consonants = 4
4	-	Program exits

C Program

```
#include <stdio.h>

void countCharacters()
{
    char str[100];
    int count = 0;
    printf("Enter a string: ");
    scanf("%[^\n]", str);

    while (str[count] != '\0')
        count++;

    printf("Number of characters = %d\n", count);
}

void printCharacters()
{
    char str[100];
    int i = 0;
    printf("Enter a string: ");
    scanf("%[^\n]", str);

    printf("Characters are:\n");
    while (str[i] != '\0')
    {
        printf("%c\n", str[i]);
        i++;
    }
}

void countVowelsConsonants()
{
    char str[100];
    int i = 0, vowels = 0, consonants = 0;

    printf("Enter a string: ");
    scanf("%[^\n]", str);

    while (str[i] != '\0')
    {
        if ((str[i] >= 'A' && str[i] <= 'Z') ||
            (str[i] >= 'a' && str[i] <= 'z'))
        {
            if (str[i]=='A' || str[i]=='E' || str[i]=='I' ||
                str[i]=='O' || str[i]=='U' || str[i]=='a' ||
                str[i]=='e' || str[i]=='i' || str[i]=='o' ||
                str[i]=='u')
                vowels++;
            else
                consonants++;
        }
        i++;
    }

    printf("Number of vowels = %d\n", vowels);
    printf("Number of consonants = %d\n", consonants);
}

int main()
{
    int choice;

    do
    {
        printf("\n----- MENU ----- \n");
        printf("1. Count characters in a string\n");
        printf("2. Print each character on a new line\n");
        printf("3. Count vowels and consonants\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1: countCharacters(); break;
            case 2: printCharacters(); break;
            case 3: countVowelsConsonants(); break;
            case 4: printf("Exiting program...\n"); break;
        }
    }
}
```

```
        default: printf("Invalid choice. Please try again.\n");
    }
} while (choice != 4);

return 0;
}
```

Execution

Terminal output of the compiled and executed program:

```
dhruvpandey@DHRUVs-MacBook-Air ~ % clang day36.c -oday36
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day36
zsh: command not found: ./day36
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day36
```

```
----- MENU -----
1. Count characters in a string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 1
Enter a string: Dhuv
Number of characters = 4
```

```
----- MENU -----
1. Count characters in a string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 2
Enter a string: DHruv
Characters are:
D
H
r
u
v
```

```
----- MENU -----
1. Count characters in a string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 3
Enter a string: Dhruv
Number of vowels = 1
Number of consonants = 4
```

```
----- MENU -----
1. Count characters in a string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 4
Exiting program...
dhruvpandey@DHRUVs-MacBook-Air ~ %
```